

Minimum Path with Resource Consumption Constraints

Given a graph $G = (V, A)$ with travel times t_{ij} associated with the edges $(i, j) \in A$, we wish to determine the minimum-time path from node s to node t , knowing, however, that passing through each node i requires a consumption r_i of a certain resource for which there is a limited availability R .

Formulate a mathematical model for this problem and the corresponding model in AMPL. Solve a small-scale instance, commenting on the results and trying to modify some data, noting how the solution changes.